Project

Bases de dados/Bases de dados e analise de informação
Department of Informatics Engineering

Objectives

- After learning how to develop an app in django in lab classes, experiment with developing your own web app

Group=2 students

Final Delivery

- You must submit your project in a zip file using Inforestudante. Do not forget to associate your work colleague during the submission process.
- The submission contents are:
  - Source code of the requested application ready to compile and execute.
  - A small report **with the functional requirements** for the application, the data model (entity-relationship model + physical model), % of work of each group member (expected should be 50% each group member), and expectation of score of each group member (0-100%).

How the functional requirements are written:
One paragraph for each requirement (may be numbered)
The requirements must be complete and accurate regarding what the intended use of the database needs
The ER diagram must solve the problem completely and accurately
The python/Django programming may be restricted to only some parts of the app when the size gets too big for the course

Grading

- REPORT (confirmed later by defense)
  - Final defense/presentation of the work.
  - Students show the app working via zoom and teacher asks questions to see if the students know how it was coded (they did it themselves)
  - The score is individual -> if a member did not work much, he has a bad score compared with colleague
  - Works with a data model without 1:m relationships or less than 7 tables are severely downscored to less than 30%
  - Works not running will score at most 10%
  - Copies of projects of other students will score 0

Resources

- Django
- Python
- Postgres

Project Description

In this project, you will be developing a part of a web app using Django. The database will have to be postgres and you will need to choose your own context and corresponding ER diagram to start with.

The students decide what web app to develop according to their desire. Whatever is chosen must be realistic, do not simplify.

Examples: online bookshop; flowers selling site; furniture selling site; …

Students will scale down the app to make sure they only have to implement a few screens, similar to guitars app.

Web app requirements:

1. Tem de ter pelo menos 5 entidades relacionadas entre si (não incluindo tabelas de ligação), com relacionamentos 1:m incluidos.
2. Tem de ter o diagrama ER e o diagrama físico.
3. Ter os mesmos elementos que tem a aplicação Guitars, incluindo mas não só:
    - Criar o seu modelo no models.py correctamente
    - Views e templates django para mostrar os elementos inseridos em cada entidade.
    - Permitir adicionar novos elementos às entidades usando o admin do django para tal.
    - Permitir adicionar e alterar elementos das entidades usando formulários / écrans desenvolvidos pelo grupo.
    - Terá menus/opções para todas as funcionalidades
    - Ter página index do django com links para cada entidade indicando o numero de elementos da tabela respectiva.
    - Os restantes elementos que vê em Guitars

Pretende-se também acomodar duas hipóteses, para servir quer alunos que apenas pretendem aprender o básico de desenvolvimento e alunos que queiram ir mais longe na aprendizagem autonoma:

Opção 2. Quero desenvolver mais a aplicação para se tornar uma webapp profissional, com html e css evoluídos, menus e toda a logica típica profissional. O projecto valerá x + y na nota da cadeira, o exame valerá z-y (ver os valores na informação geral da cadeira).